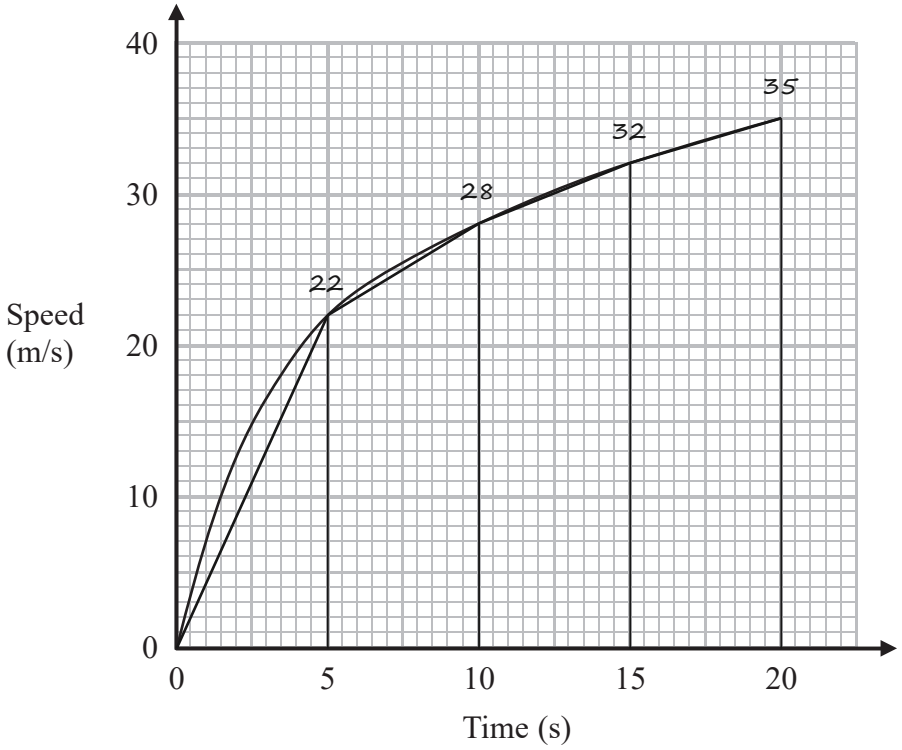
15 The graph shows the speed of a car, in metres per second, during the first 20 seconds of a journey.



(a) Work out an estimate for the distance the car travelled in the first 20 seconds.
Use 4 strips of equal width.

Four equal widths: 20 ÷ 4 = 5 seconds.

Read speeds ≈ 0, 22, 28, 32, 35 m/s.

For each trapezium, A = ½(a + b)h.

1 mark
Use a trapezium area

$A_1 = ½(0 + 22) \times 5 = 55$

$A_2 = ½(22 + 28) \times 5 = 125$

$A_3 = ½(28 + 32) \times 5 = 150$

$A_4 = ½(32 + 35) \times 5 = 167.5$

1 mark
Use all four equal strips

Distance ≈ 55 + 125 + 150 + 167.5 = 497.5 m

1 mark
Correct estimate within the graph-reading range

497.5 metres

(3)

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(b) Is your answer to part (a) an underestimate or an overestimate of the actual distance the car travelled in the first 20 seconds?
Give a reason for your answer.

Underestimate: the straight tops of the trapezia lie below the curve.

They leave out some of the area under the speed–time graph.

1 mark
Link the missing area to the distance estimate

(1)

(Total for Question 15 is 4 marks)

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